

Do We Need Stratified Cross-Validation to Compare Classification Algorithms?

A Repeated 5-Fold Study Across 13 OpenML Datasets

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Abstract

Cross-validation is routinely used to compare classifiers, yet the construction of the folds may influence the estimated performance and, potentially, the identity of the better algorithm. This paper studies whether stratified 5-fold cross-validation changes the comparison between Logistic Regression and a Decision Tree on 13 OpenML classification datasets. The benchmark contains between 150 and 48,842 observations and minority-class shares from 2.3% to 44.5%. Regular shuffled K-Fold and shuffled Stratified K-Fold are each repeated ten times with matched random seeds. Preprocessing is fitted inside every training fold, and performance is evaluated with Macro- F_1 . Across 2,600 model fits, stratification reduces the standard deviation of the smallest class share across folds by a median of 93.4%. However, its average score effect is only +0.00118 for Logistic Regression and +0.00147 for the Decision Tree; neither effect is statistically significant. The change in the between-algorithm performance gap is also non-significant (Wilcoxon $p = 0.946$), and no dataset changes its winning algorithm. Thus, stratification is an effective safeguard for fold composition, but it is neither a universal performance booster nor a determinant of the algorithm ranking in this benchmark.

Keywords: cross-validation, stratification, imbalanced classification, algorithm comparison, reproducibility, OpenML

1 Introduction

Model evaluation is part of the scientific claim made by an empirical machine-learning study. When a fixed train/test split is too variable or wastes scarce data, cross-validation (CV) is commonly used to estimate generalization performance. In K -fold CV, the observations are partitioned into K mutually exclusive folds; each fold is used once for evaluation while the others are used for training. Repeating the procedure with different shuffles provides several estimates and makes the dependence on one random partition visible.

For classification, ordinary shuffled K-Fold does not constrain the class distribution in an individual fold. On an imbalanced dataset, a test fold can therefore contain very few observations from the minority class. This is especially relevant for a class-sensitive metric such as Macro- F_1 , which gives equal weight to each class. Stratified K-Fold addresses this mechanical problem by approximately preserving the global label proportions in every fold. It is often recommended as a safe default, but a more stable label composition does not automatically imply a larger, less biased, or statistically different performance estimate (scikit-learn developers, 2026).

The scientific concern in this project is comparative rather than predictive. If two reasonable evaluation protocols reverse the ranking of two algorithms, researchers can draw incompatible conclusions from the same data. The research question is therefore:

Do we need stratified cross-validation to obtain a stable comparative conclusion between classification algorithms on balanced and imbalanced datasets?

The study compares Logistic Regression (LR), which represents a regularized linear decision boundary, and an unpruned Decision Tree (DT), which represents a nonlinear, high-variance learner. The primary outcome is the LR–DT score gap under each CV protocol. The null hypothesis H_0 states that the dataset-level change in this gap is centered at zero and that the protocol does not systematically change the algorithm comparison. The alternative H_1 states that the gap changes systematically and/or changes enough to reverse the winner on at least one dataset. Two-sided paired tests use a significance level of $\alpha = 0.05$; practical relevance is evaluated from effect magnitudes and ranking reversals as well as p -values.

The paper makes three contributions:

- a repeated, paired evaluation on 13 heterogeneous datasets rather than a conclusion based on one example;
- an operational analysis of score shifts, estimate variability, fold composition, comparison-gap changes, and ranking flips; and
- a leakage-safe and reproducible implementation with fixed OpenML IDs, fixed random seeds, explicit software versions, and compute reporting.

2 Background

2.1 Regular and stratified cross-validation

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ be a labeled dataset. In shuffled K -fold CV, a random permutation of $\mathcal{D}$ is partitioned into K disjoint test folds. Each observation appears in exactly one test fold per repeat. The procedure aims to distribute observations evenly, but it does not constrain the distribution of y .

Stratified K -Fold performs the partition conditionally on the class label. For each class, its observations are distributed across folds so that the class proportions are approximately preserved. This prevents pathological splits with a missing or extremely small class, provided the number of examples is sufficient. Stratification controls only the marginal distribution of the target. It does not balance feature distributions, difficult subgroups, duplicates, groups, or temporal drift. Consequently, it is an engineering solution to one source of split variability, not a general solution to all forms of sampling bias.

2.2 Class imbalance and the evaluation metric

Class imbalance can interact with sample size, concept complexity, and the inductive bias of a classifier (Japkowicz and Stephen, 2002). Accuracy can conceal poor minority-class performance because the majority class dominates the count of correct predictions. This

study therefore uses Macro- F_1 . For C classes, class-wise precision P_c , and recall R_c , it is

$$F_{1,\text{macro}} = \frac{1}{C} \sum_{c=1}^C \frac{2P_c R_c}{P_c + R_c}. \quad (1)$$

Each class contributes equally, irrespective of its prevalence. When a prediction contains no positive examples for a class, the implementation uses a zero-division value of 0. This makes the metric definition explicit and avoids undefined fold results.

2.3 Comparing algorithms across datasets

Fold-level observations from one dataset are not independent: their training sets overlap and the same observations are reused across repeats. Treating all folds as independent would exaggerate the effective sample size. Following recommendations for multi-dataset comparisons (Demšar, 2006), datasets are the units of the primary statistical analysis. The paired Wilcoxon signed-rank test is used because only 13 dataset-level effects are available and normality cannot be assumed. A paired t -test is reported as a sensitivity analysis, while effect sizes and rank reversals provide the practical interpretation.

3 Relevant Work

Early empirical work showed that the choice of resampling procedure affects error estimation and model selection. Kohavi (1995) compared cross-validation and bootstrap variants and emphasized the trade-off between bias and variance. The present study asks a narrower question: whether label-preserving fold construction, while keeping the number of folds and repeats fixed, changes an algorithm comparison.

Demšar (2006) developed practical guidance for statistical comparisons of classifiers over multiple datasets. Their central lesson—pairing results at the dataset level and using non-parametric procedures when appropriate—motivates the main test in this paper. The current experiment differs from broad classifier benchmarks because the two learners are fixed and the manipulated factor is the CV splitter.

The empirical infrastructure follows the reproducibility goals of OpenML (Vanschoren et al., 2014) and the estimator and pipeline interfaces of scikit-learn (Pedregosa et al., 2011). The initial project proposal used Blood Transfusion and Iris as illustrative examples. Those two datasets suggested small differences, but they span neither the severe imbalance nor the size variation needed for a broader conclusion. The final benchmark therefore expands the design to 13 datasets, ten repeated splits, fold-local preprocessing, and dataset-level paired tests.

4 Methodology and Experimental Setup

4.1 Benchmark datasets

The datasets were fixed in the project proposal and fetched by OpenML data ID, preventing substitution with similarly named variants. Table 1 summarizes the suite. It spans small and medium sample sizes, numeric and categorical features, binary and multiclass targets, and a wide imbalance range. Iris is the only three-class dataset; all others are binary. PC1

Table 1: Benchmark datasets. n is the number of observations, p the number of input features, and “Minority” the smallest class share.

Dataset	OpenML ID	n	p	Minority share
Blood Transfusion	1464	748	4	23.8%
Adult	1590	48,842	14	23.9%
QSAR Biodegradation	1494	1,055	41	33.7%
Spambase	44	4,601	57	39.4%
Iris	61	150	4	33.3%
German Credit	31	1,000	20	30.0%
KC1	1067	2,109	21	15.5%
PC1	1068	1,109	21	6.9%
ILPD	1480	583	10	28.6%
Diabetes	37	768	8	34.9%
Credit Approval	29	690	15	44.5%
Mammography	310	11,183	6	2.3%
Phoneme	1489	5,404	5	29.3%

and Mammography are the most severe cases, with minority-class shares of 6.9% and 2.3%, respectively.

This is a deliberately heterogeneous benchmark, not a random sample of all classification tasks. Inference therefore applies to the selected suite. The breadth is nevertheless important because a single balanced dataset would not stress the property that stratification is designed to control.

4.2 Experimental factors and models

For every dataset, two splitters are compared:

1. REGULAR-CV: shuffled 5-fold K-Fold;
2. STRATIFIED-CV: shuffled 5-fold Stratified K-Fold.

Both procedures are repeated ten times. Repeat r uses seed $235975 + r$, and the same sequence of repeat seeds is used for both protocols and algorithms. This matching removes an avoidable source of noise while preserving the intended difference in fold construction.

The classifiers are an L2-regularized LR fitted with the `lbfgs` solver and at most 2,000 iterations, and an unpruned DT. The tree’s random seed is fixed to the repeat seed. No class weighting, resampling, feature selection, or hyperparameter tuning is performed. These exclusions are deliberate: they keep the splitter as the only intended experimental difference and avoid a second model-selection layer.

4.3 Leakage-safe preprocessing

Preprocessing is contained in a pipeline and refitted on every training fold. Numeric variables are imputed with the training-fold median and standardized. Categorical variables are imputed with the training-fold mode and one-hot encoded with unknown categories ignored. The fitted transformations are then applied to the corresponding held-out fold.

This design prevents information from the test fold from influencing imputation values, scaling parameters, or the category vocabulary.

Each split is reused for LR and DT, preserving a paired comparison within a protocol. The prediction on the held-out fold is scored with Macro- F_1 . Aggregating $10 \times 5 = 50$ fold scores produces one mean per dataset–protocol–algorithm combination. With 13 datasets, two protocols, two algorithms, ten repeats, and five folds, the full experiment contains

$$13 \times 2 \times 2 \times 10 \times 5 = 2,600 \quad (2)$$

model fits.

The resulting aggregation hierarchy is fold $\rightarrow$ repeat $\rightarrow$ protocol $\rightarrow$ dataset. Let $s_{d,p,a,r,k}$ denote a fold score for dataset d , protocol p , algorithm a , repeat r , and fold k . The reported mean is

$$\bar{F}_{1,d,p,a} = \frac{1}{10} \sum_{r=1}^{10} \left(\frac{1}{5} \sum_{k=1}^5 s_{d,p,a,r,k} \right). \quad (3)$$

This expression makes explicit which dimensions are averaged and prevents a fold-level test from being mistaken for a dataset-level comparison.

4.4 Outcomes and hypothesis tests

For dataset d and protocol p , define the algorithm-comparison gap

$$G_{d,p} = \bar{F}_{1,d,p,\text{LR}} - \bar{F}_{1,d,p,\text{DT}}, \quad \Delta G_d = G_{d,\text{Stratified}} - G_{d,\text{Regular}}. \quad (4)$$

A positive gap favors LR, a negative gap favors DT, and a ranking flip occurs when the signs under the two protocols differ. The primary test applies a two-sided Wilcoxon signed-rank test to the 13 values of ΔG_d . A paired t -test is reported as a sensitivity check.

Secondary outcomes are: (i) the stratified-minus-regular mean Macro- F_1 separately for each model; (ii) the standard deviation of the ten repeat-level mean scores, analyzed over 26 dataset–algorithm pairs; (iii) the standard deviation of the smallest class share across the five test folds; and (iv) the Spearman association between dataset minority share and the average protocol score effect. These outcomes distinguish a mechanical improvement in fold construction from a performance change.

5 Implementation Details

5.1 Software and parallel execution

The experiment uses Python 3.12.13, scikit-learn 1.8.0, NumPy 2.3.5, pandas 2.2.3, and SciPy 1.17.0. OpenML provides the fixed dataset identities and automated retrieval. scikit-learn implements the splitters, preprocessing pipelines, estimators, and metric; pandas and NumPy organize the results; SciPy performs the paired Wilcoxon, paired t , and Spearman tests.

Independent dataset–repeat–protocol tasks are executed with four parallel workers. Within a task, the same folds are used for both algorithms. The machine exposes nine logical CPUs and runs x86_64 Linux. The complete run takes 105.8 seconds of wall-clock time; summed fit-and-score time is 376.2 seconds. The difference reflects parallel execution and the fact that the summed time counts concurrent work separately.

5.2 Code and reproducibility artifacts

Reproducibility controls include fixed OpenML IDs, a documented base seed, matched seeds across protocols, fold-local preprocessing, explicit model configurations, and retention of fold-level scores and timings. The complete reproduction script, reported result tables, figure-generation code, environment requirements, and JMLR source are available at <https://github.com/ibrahem2001ab/stratified-cv-comparison>. The repository README documents the commands for rerunning the 2,600 model fits and rebuilding this paper.

Listing 1 shows the central evaluation logic in simplified form. The actual repository version also records timing, fold identifiers, dataset metadata, aggregate tables, statistical tests, and the software environment.

Listing 1: Core leakage-safe evaluation loop. Both algorithms reuse the same train/test indices.

```
for train_idx, test_idx in splitter.split(X, y):
    for name, estimator in models(seed).items():
        estimator.fit(X.iloc[train_idx], y.iloc[train_idx])
        prediction = estimator.predict(X.iloc[test_idx])
        score = f1_score(
            y.iloc[test_idx], prediction,
            average="macro", zero_division=0,
        )
```

The repository separates the recorded results from newly reproduced outputs. Published summary files remain under **results/**, whereas a new run writes to **results/reproduced/**. Three offline smoke tests check the declared fit count, mixed numeric/categorical preprocessing, and the expected reduction in class-share variation on synthetic imbalanced data. A continuous-integration workflow runs these checks whenever the repository is updated.

6 Results

6.1 Stratification stabilizes fold composition

The direct effect of stratification is large and consistent: the standard deviation of the smallest class share across the five test folds falls by a median of 93.4% across datasets. German Credit and Iris have identical smallest-class shares in all five stratified test folds, while Mammography is nearly exact despite its 2.3% minority class. This confirms that the implementation behaves as intended and that the selected benchmark includes conditions in which unconstrained splitting can visibly change test-fold composition.

The class-balance range of the benchmark is shown in Figure 1. The figure is descriptive rather than inferential: it identifies the stress cases for which a class-preserving splitter has the clearest mechanical motivation.

6.2 Mean score effects are small

Table 2 reports the paired statistical tests. The average stratified-minus-regular effect is +0.00118 for LR and +0.00147 for DT. Neither the Wilcoxon test nor the paired *t*-test

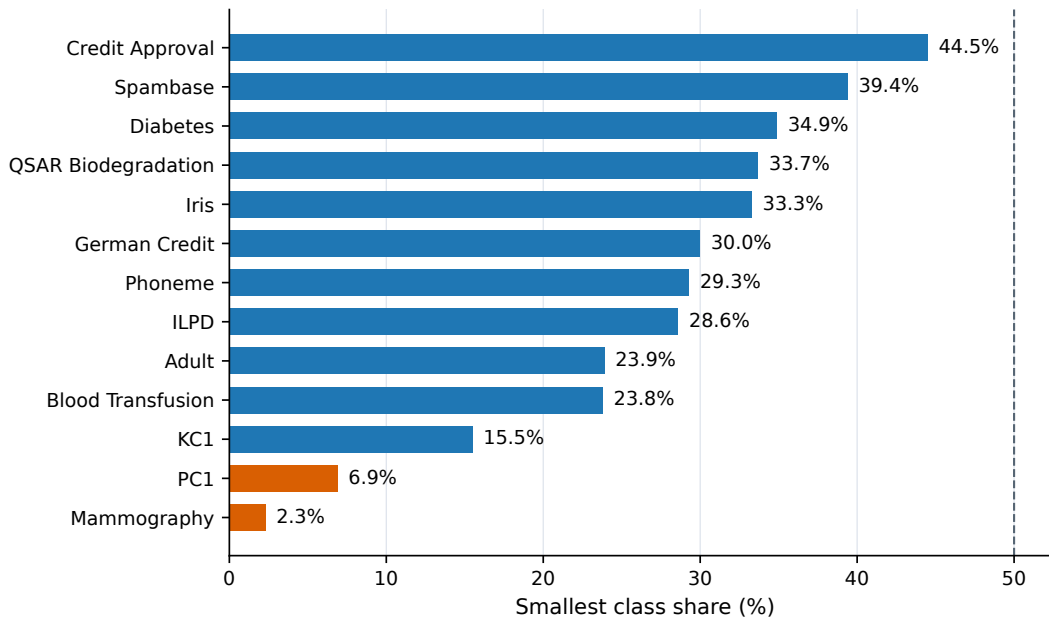


Figure 1: Smallest class share for the 13 OpenML datasets. The dashed line marks the balanced 50% value for binary data; Iris is a balanced three-class dataset at 33.3%.

Table 2: Paired effects of STRATIFIED-CV minus REGULAR-CV. Score tests use 13 datasets. The variability test uses 26 dataset–algorithm pairs.

Outcome	Mean	Median	Wilcoxon p	Paired t p
LR mean Macro- F_1	+0.00118	+0.00033	0.635	0.323
DT mean Macro- F_1	+0.00147	+0.00127	0.414	0.437
Repeat-to-repeat score SD	−0.00094	−0.00015	0.247	0.147

rejects a zero dataset-level effect. Effects do not point consistently in one direction: the largest gains are +0.0110 for LR on PC1 and +0.0116 for DT on QSAR Biodegradation, while the largest loss is −0.0101 for DT on Mammography. Thus, stabilizing class allocation does not produce a uniform increase in Macro- F_1 .

The repeat-to-repeat standard deviation decreases by only 0.00094 on average and 0.00015 at the median. Its Wilcoxon p -value is 0.247. Therefore, a more predictable label share within a fold should not be conflated with a reliably lower variability of the final repeated-CV score.

6.3 The algorithm ranking is unchanged

Table 3 presents the primary outcome. Seven datasets favor LR under both protocols, and six favor DT under both. None of the 13 signs changes. Figure 2 visualizes the same

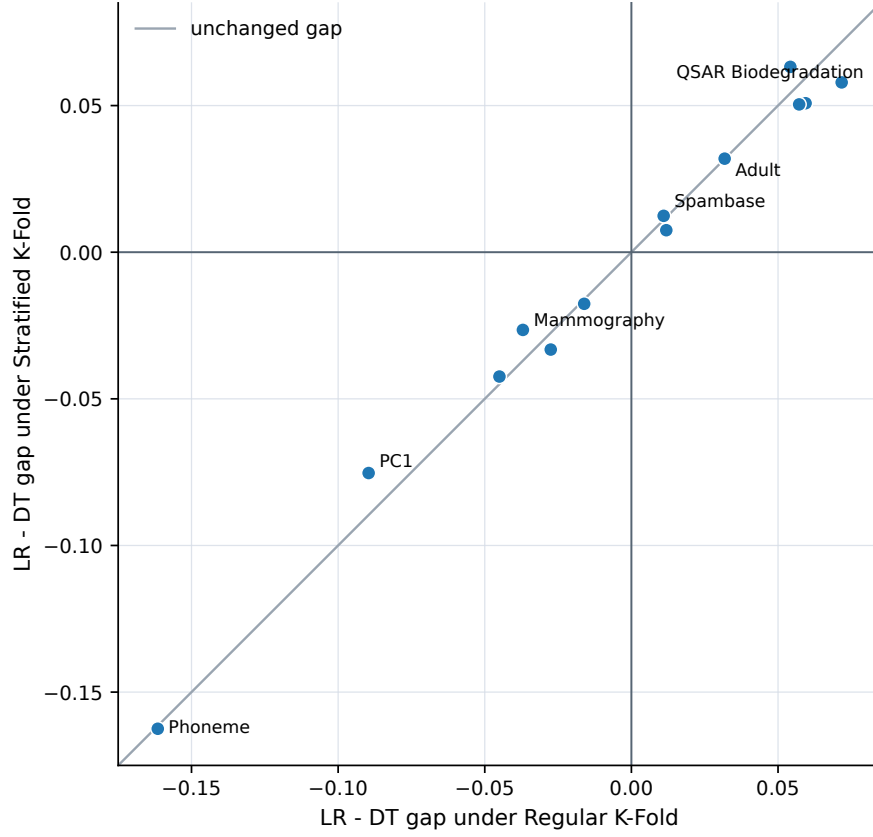


Figure 2: LR–DT Macro- F_1 gap under regular and stratified 5-fold CV. Positive values favor LR; negative values favor DT. A ranking flip would fall in an off-diagonal quadrant. All points remain in a same-winner quadrant.

result: every dataset stays in a quadrant that represents the same winner under regular and stratified CV.

Across datasets, ΔG has mean -0.00029 , median -0.00094 , and median absolute magnitude 0.00568 . The primary two-sided Wilcoxon signed-rank test does not reject H_0 ($W = 44$, $p = 0.946$), and the paired t -test agrees ($p = 0.898$). The largest absolute gap change is 0.0143 on PC1; the gap remains negative, so DT still ranks higher. Numerically different estimates therefore do not imply a substantively different comparison.

An exploratory Spearman test finds no association between minority-class share and the mean protocol score effect across the two algorithms ($\rho = 0.082$, $p = 0.789$). Within this suite, more severe imbalance does not imply a larger score gain from stratification. The test has low power with 13 observations and should not be interpreted as proof that no relationship exists in a broader population.

Table 3: Algorithm-comparison gaps $G = \bar{F}_{1,\text{LR}} - \bar{F}_{1,\text{DT}}$. Positive values favor LR. ΔG is stratified minus regular.

Dataset	Regular gap	Stratified gap	ΔG
Adult	+.0318	+.0319	+.0001
Blood Transfusion	-.0450	-.0424	+.0026
Credit Approval	+.0572	+.0504	-.0067
Diabetes	+.0542	+.0632	+.0089
German Credit	+.0594	+.0508	-.0086
ILPD	-.0161	-.0176	-.0014
Iris	+.0119	+.0075	-.0044
KC1	-.0275	-.0332	-.0057
Mammography	-.0370	-.0265	+.0105
PC1	-.0896	-.0753	+.0143
Phoneme	-.1615	-.1625	-.0009
QSAR Biodegradation	+.0717	+.0579	-.0138
Spambase	+.0110	+.0124	+.0014

7 Discussion

7.1 Why composition stability does not guarantee a score gain

The evidence separates two meanings of “stability.” First, fold-composition stability improves sharply: stratification makes the proportion of the smallest class predictable. Second, inferential stability—the variability and central value of the performance comparison—changes little. These findings are compatible because label prevalence is only one determinant of a fold’s difficulty. Conditional feature distributions, rare subgroups, noisy labels, duplicates, and decision-boundary geometry remain uncontrolled.

For the benchmark studied here, the primary scientific conclusion is stable. Stratification changes some numeric estimates by roughly one Macro- F_1 percentage point, but the LR–DT gap remains on the same side of zero for every dataset. The absence of a ranking reversal is more directly relevant to the research question than a small change in one model’s isolated score.

The result does not justify abandoning stratification. A class-preserving splitter is a low-cost safeguard when observations are exchangeable and the target is imbalanced. It reduces the risk of folds with a sparse or absent class and makes each fold more representative with respect to the target marginal. The appropriate recommendation is therefore to use stratification by default in ordinary imbalanced classification, while describing it as a protection against pathological label allocation rather than as a performance-improvement method.

7.2 Practical protocol selection

The splitter must respect the data-generating process. Group-aware CV is required for repeated patients, users, or devices, and time-aware evaluation is required for chronological prediction. For exchangeable imbalanced samples, Stratified K-Fold is a suitable starting

point; the split should be repeated with documented seeds and algorithms should be compared with paired dataset-level effects.

8 AI Usage Declaration

Generative AI assisted with code scaffolding, LaTeX formatting, language editing, and visual checks. Automated tools supported data processing, experiment execution, aggregation, and chart generation. The author selected the question, data, models, metric, and design, and verified the code and numerical claims against recorded outputs. AI was not treated as an empirical source. It aided learning about reproducible paired experiments and the difference between fold-composition and performance stability.

9 Limitations

Construct validity. The conclusions cover two untuned algorithms and Macro- F_1 . Ensembles, tuned pipelines, calibration measures, PR-AUC, or application-specific costs may respond differently to the split protocol.

External validity. The 13 OpenML datasets are heterogeneous but not a random sample of all classification tasks. The inference describes this benchmark; rarer classes may require more datasets, repeats, or repeated holdout designs.

Internal validity. The benchmark contains no grouped or time-dependent task. The recommendation applies only to exchangeable observations; group and temporal boundaries take priority over ordinary stratification.

Conclusion validity. With 13 datasets, the tests have limited power for very small effects. Failure to reject H_0 is not proof of equivalence; the descriptive result is no ranking reversal and a median absolute gap change of 0.00568.

10 Conclusion and Future Work

Under repeated shuffled 5-fold CV on the 13 specified OpenML datasets, stratification is not required to preserve the comparative conclusion between LR and DT. No dataset changes its winning algorithm, and the dataset-level change in the comparison gap is non-significant ($p = 0.946$). The null hypothesis is therefore not rejected.

Stratification nevertheless has a clear operational benefit: it reduces fold-to-fold class-composition variability by a median of 93.4%. The practical conclusion is concise: stratify for robustness when samples are exchangeable, but compare algorithms using repeated, paired evidence and do not present stratification as a guarantee of higher expected performance.

Future work should test tuned classifiers, PR-AUC and calibration, more datasets and repeats, and splitters that combine label balance with group or temporal constraints. It should also test whether stratification matters more for model selection than for fixed-model comparison.

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